Vishal Murugadhas

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Education

Sardar Vallabhbhai National Institute of Technology **2024–2028**
B.Tech, Electronics and Communication Engineering *CGPA: 8.95*

The Indian School, Bahrain **2022–2024**
CBSE Class XII: 92.2% | Class X: 84.2%

Technical Skills

Languages: C++, Python, Java, C
ML/AI: Scikit-learn, XGBoost, Pandas, NumPy, YOLOv8, IndicBERT, Anomaly Detection (IsolationForest), Time-Series Modeling
Web & Backend: React, Next.js, Node.js, Express.js, FastAPI
Data & Infra: Kafka, Flink, MongoDB, Firebase, Docker, Git/GitHub
Simulation: SimPy, SUMO, TraCI
Hardware: Verilog, RISC-V ISA, FPGA (Xilinx/Vivado), Digital Logic Design

Projects

PHANTOM – Privileged-Access Insider-Threat Detection Platform ***(2nd Place, 450+ teams)*** GitHub
- Led NLP and data-pipeline engineering for a 3-person team at the Bank of Baroda × IIT Gandhinagar PSB Hackathon; contributed to a Bayesian-weighted Dynamic Insider Threat Score combining 4 detection engines.
- Built the Negative Access Profiler and an IndicBERT-based Justification NLP Scorer, which reached **98.33% accuracy**.
- Designed the synthetic telemetry generator and Kafka/Flink streaming pipeline that produced **977,705 access events** across **50 employees**, yielding **4,500 employee-day records** with 37 behavioral features.

AegisAI – AI-Powered Autonomous Emergency Response System GitHub
- Built an autonomous emergency-response platform combining **25 AI-powered CCTV nodes**, multi-modal computer vision, and FastAPI dispatch orchestration to coordinate **15 drones across 5 dispatch hubs**.
- Designed a multi-stage detection pipeline (YOLOv8-pose, YOLOv8s, ByteTrack, Farneback Optical Flow, IsolationForest) for falls, fires, accidents, and crowd anomalies; achieved **98.1% weapon-detection precision** at up to **65 FPS** on an RTX 4090.
- Implemented severity/priority scoring with **200m spatial deduplication** and no-fly-zone-aware routing, reaching **<800ms end-to-end dispatch latency** in benchmarked simulations.

StockSense India – AI-Powered Market Predictor GitHub | Live Demo
- Built a full-stack market-intelligence platform for **NSE/BSE equities**, forecasting next-day direction with a leakage-resistant Logistic Regression pipeline trained on **5+ years of OHLCV data**.
- Engineered a 14-feature time-series vector (return lags, rolling volatility, momentum, volume velocity); achieved **52–68.5% out-of-sample directional accuracy** using chronological train/test splits.

Leadership

Google Developer Groups on Campus (GDGC), SVNIT **2025–Present**
AI/ML Lead
- Lead the AI/ML vertical of the club, guiding members on projects and technical learning in machine learning and AI.

MINDBEND, SVNIT **Jan 2025**
Organizing Team
- Planned and executed a university-level MUN event with 50+ participants, managing outreach and cross-team communication.

Achievements

- **2nd Place**, Bank of Baroda × IIT Gandhinagar PSB Hackathon 2026, among **450+ teams**; awarded a **Rs. 3 Lakh** prize for PHANTOM.
- **Winner**, Inspiron 5.0 Hackathon (COEP Pune).
- Developed the official website for **Prism**, the annual sports event at SVNIT.
- Solved 150+ LeetCode problems; ranked in the top 15% of the class by CGPA.